

Classifying Survey Comments – Keyword matching vs Td-Idf vs Setfit

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July 2026

Overview

- This session stems from my recent project work on customer survey classification for Customer Insight Team
- Every customer satisfaction survey generates hundreds of free-text comments that need a theme
- Keyword rules are natural first attempt, simple to write, easy to explain
- However, people rarely use the exact words, indirect comments get missed
- Machine Learning models learns the underlying patterns instead of relying on exact wording

Our session today



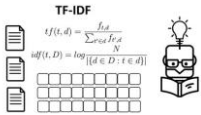
Dataset

400 synthetic workplace comments generated by copilot across 4 themes



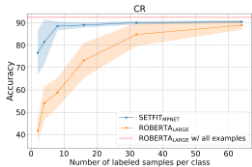
Keyword matching

Rule-based classification



ML Classifier

TF-IDF + sckit-learn : Learns patterns from examples



SetFit

Sentence level understand – what I used for recent project



Comparison

Compare accuracy across all three approaches



Wrap-up

Review key takeaways